Differential Equations II

1. First-order ordinary differential equations

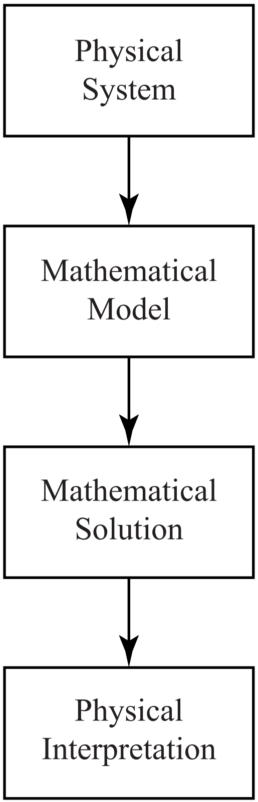
Modelling

There are three steps here

1. Formulate a model
2. Solve the model
3. Interpret the results

Pure mathematics focuses primarily on the second step and doesn't concern itself with the first and third steps.

Applied mathematics is all three steps.



Modelling

Often physical models will involve a derivative.

Address situations where the mathematical model is a differential equation: typically we seek a function $y(x)$ (or $y(t)$) and then modelling gives us an equation involving its derivative dy/dx, often denoted y' or $\dot{y}$.

Modelling (continued)

This could be in mechanical models which involve relationships between force, mass and acceleration. Like a parachutist gently falling to earth.

It could be population models where there are relationships between say the size of a population and the effect that has on another population. Such a model of a predator-prey relationship in biology.

Or it could involve the quantum interactions between sub-atomic particles, electro-magnetism or any of a large number of fields.

Basic Concepts

Definition

A **first-order ordinary differential equation (ODE)** is an equation involving only one variable x, one function $y(x)$ and its derivative dy/dx in either

$$\text{an } \textit{implicit form:} \quad F(x, y, y') = 0, \tag{1}$$

$$\text{or the } \textit{explicit form:} \quad y' = f(x, y). \tag{2}$$

Unpack this definition

Example 1: Carbon dating - how old is this painting?

ABC News, 25 February 2021:

Oldest-known rock art in Australia is 17,300-year-old kangaroo in the Kimberley, wasp nests show

https://www.abc.net.au/news/science/2021-02-23/oldest-rock-art-australia-kimberley-kangaroo-mud-wasp-nest/13173792

Example 1: Physical system – carbon dating

Observations

- ^{14}C is present in all living things to the same proportion.
- In the Kimberley rock art, approximately 12.3% of the ^{14}C remains.
- The decay in a small period of time Δt is proportional to the time scale Δt and the amount of ^{14}C present.
- the half-life of ^{14}C is 5,730 years.

Example 1: Let's turn this information into a model!

Basic Concepts

Definition

A function $y = h(x)$ is called a **solution**, or a **particular solution**, if it satisfies the ODE on some open interval. A solution with an arbitrary constant c, $y = h(x; c)$, is called a **general solution**.

Definition

An **initial value problem** (IVP) is formed by an ODE together with an **initial condition** $y(x_0) = y_0$ for the solution to satisfy.

Example 1: Solution – carbon dating

Our ODE is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -ky.$$

We know from first year that the solution of this equation is

$$y(t) = ce^{-kt}.$$

This is a general solution to the ODE.

Now $y(0) = c$, so c has the physical interpretation of being the amount of ^{14}C present at the time that the cave art was produced. Let's call this quantity $c = 1$.

We also know from the half-life that $y(5730) = 0.5$, which gives us the value of $k \approx 1.20968 \times 10^{-4}$. So we have

$$y(t) = e^{-1.20968 \times 10^{-4} t}.$$

This is the particular solution to the IVP.

Example 1: Interpretation – carbon dating

So solving for $y(t_1) = 0.123$ leads to $t_1 \approx 17{,}300$, this is the approximate age of the rock art in the ABC story.

Separable ODEs

Definition

An ODE is called **separable** if it can be put in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{f(x)}{g(y)}.$$

Consider the IVP for $y(x)$:

$$y' = -2xy, \quad \text{subject to } y(0) = 1.8.$$

Example 2: Solve this IVP (solve separable ODE, then apply initial condition)

Example 3 – a box of water

Consider the flow of water from a box of water, with a tap at the bottom. The base of the box is 20 cm by 20 cm. Its height is 25 cm. The tap opening is 1 cm by 0.4 cm. The initial height of the water is the full 25 cm.

Starting with a full box, we are interested in finding the height of water inside at any time t.

If we hold the tap open, how long will it take for the box to completely empty?

Example 3 – a box of water

Physical information – Torricelli's law

Under the influence of gravity the outflowing water has velocity

$$v(t) = 0.600\sqrt{2gh(t)}.$$

Modelling

Over a short time Δt the volume of outflowing water ΔV is

$$\Delta V = Av\,\Delta t,$$

where A is the cross-sectional area of the tap. The change of volume of water in the tank is

$$\Delta V^{\star} = -B\,\Delta h,$$

where B is the cross-sectional area of the box.

Example 3 – a box of water

Assuming water volume is conserved these two quantities must be equal so

$$-B\,\Delta h = Av\,\Delta t. \quad \Rightarrow \quad \frac{\Delta h}{\Delta t} = -\frac{A}{B}v.$$

So taking the limit at $\Delta t \to 0$ and substituting Torricelli's law for v we have a first order ODE.

$$\frac{\mathrm{d}h}{\mathrm{d}t} = -\frac{A}{B}0.600\sqrt{2gh(t)}.$$

Example 3: Solve the equation, and find out when the box empties

Example 3 – a box of water

Interpretation

We can use MATLAB produce a plot

Example 3: Make a plot (`Ex3_BoxOfWater.mlx`)

Is Torricelli's law examinable?

In exercises and examinations you will be given sufficient observational/experimental information to do the modelling using common steps.

For example, you do not have to remember Torricelli's law and similar, just how to use similar information.

Linear ODEs

Definition

A first order ODE is said to be **linear** if it can be written

$$y' + p(x)y = r(x), \qquad (3)$$

otherwise it is **nonlinear**. If the function on the right-hand side term is $r(x) = 0$, then it is also called **homogeneous**, otherwise it is **nonhomogeneous**.

Solving linear ODEs

Linear ODEs of the form

$$y' + p(x)y = r(x), \tag{4}$$

can be solved by multiplication through by an integrating factor

$$F(x) = \exp\left[\int p(x)\,dx\right].$$

This leads to the general solution

$$\begin{aligned} y(x) &= \frac{1}{F(x)}\int F(x)r(x)\,dx + \frac{c}{F(x)} \\ &= \text{response to forcing} + \text{response to initial value.} \end{aligned}$$

Hopefully we can integrate $F(x)r(x)$.

Example 4 - Integrating factor

Solve the IVP

$$y' + y\tan x = \sin 2x, \quad \text{such that} \quad y(0) = 1.$$

Example 4: Solve this IVP (integrating factor, apply initial condition)

Example 5: Hormone level - Observations

- Hormone levels in the blood vary with time
- Hormones are added by a 24 hour sinusoidal input controlled by the thyroid
- Hormones are removed by a continuous process proportional to the current level

Example 5: Hormone level - Model

Let $y(t)$ be the hormone level at time t.

The replacement rate is $A + B\cos\omega t$, where A is the mean rate, B is the amplitude of the sinusoidal variation and ω is the frequency, in this case $\pi/12$.

The removal rate is $\kappa y(t)$, where κ denotes the efficiency of the removal of hormone.

So the overall change of hormone level $=$ In – Out, thus

$$\frac{dy}{dt} = A + B\cos\omega t - \kappa y.$$

Let's further assume that we have an initial reading at time $t = 0$, which is $y(0) = y_0$.

Example 5: Hormone level - Solution

In standard form the IVP is

$$y' + \kappa y = A + B\cos\omega t, \quad \text{such that} \quad y(0) = y_0.$$

Example 5: Solve the IVP (integrating factor, apply initial condition)

Example 5: Hormone level - Interpretation

As time increases the term $e^{-\kappa t}$ reduces to zero.

The other part of the solution is called the **periodic steady-state solution** because it consists of constant and periodic terms that persist. The full solution is also called the **transient-state solution** because it includes this transition from the initial state to the steady-state.

To see this more clearly, let's plot the solution in MATLAB.

Example 5: Plot the solution (`Ex5_HormoneLevel.mlx`)

Using computer algebra

Computer algebra systems can solve many routine ODE.

You may have already tried this out with Wolfram Alpha, for example.

In this course we will use MATLAB to find algebraic solutions to ODEs.

[Note that MATLAB can also be used to find approximate computational solutions to ODEs that do not have an analytic solution (more on this later).]

Examples 6: Using computer algebra to solve ODEs in MATLAB

Existence and uniqueness

In all the initial value problems so far, we have taken two things for granted:

1. that a solutions **exists**
2. that only one solution is possible, ie. it is **unique**

Is this always the case?

Example 7: Consider the ODE

$$-xy' + y = 0, \quad \text{such that} \quad y(x_0) = y_0,$$

for various initial conditions like $y(1) = 2$, $y(0) = 2$ and $y(0) = 0$.

Example 7: Solve this problem for different initial values

Existence and uniqueness theorem

Theorem

Consider the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0. \tag{5}$$

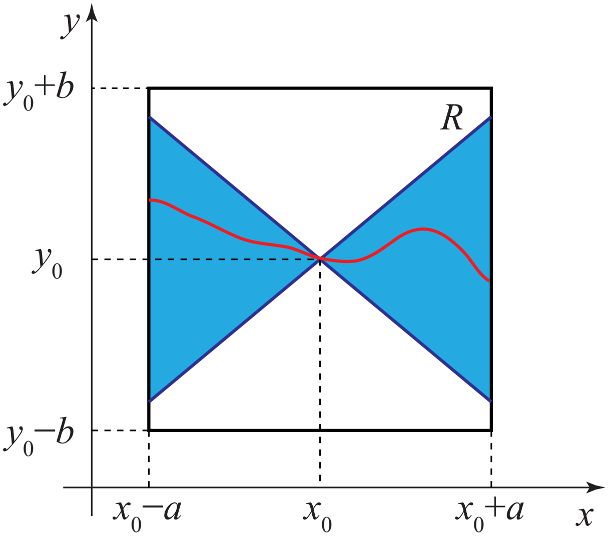
If, in some rectangle $R := \{(x, y) : |x - x_0| < a, |y - y_0| < b\}$*, the function* $f(x, y)$ *is continuous and bounded,* $|f| \leqslant K$*, then the IVP* (5) *has* **at least one** *solution* $y(x)$ *for* $|x - x_0| < \min(a, b/K)$*.*

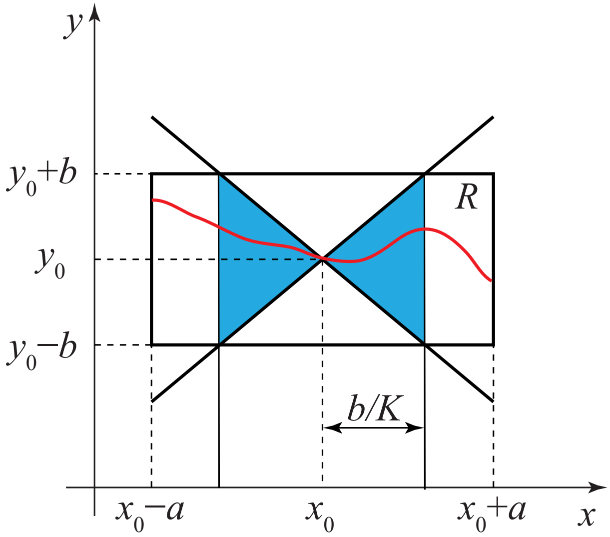
Further, if also $f_y = \partial f / \partial y$ *is continuous and bounded in the rectangle* R*,* $|f_y| \leqslant M$*, then then* IVP (5) *has a* **unique** *solution* $y(x)$ *for* $|x - x_0| < \min(a, b/K)$*.*

Proof.

See PURE MTH 3002 - Topology and Analysis III. $\square$

Existence and uniqueness picture





Existence and uniqueness

Example 8: Consider the initial value problem

$$y' = 1 + y^2, \quad \text{with} \quad y(0) = 0,$$

in the rectangle $|x| < 5$, $|y| < 3$.

Example 8: Apply the theorem

Existence and uniqueness

Example 9: The IVP

$$y' = \sqrt{|y|}, \quad \text{with} \quad y(0) = 0,$$

has the two solutions

$$y = 0, \quad \text{and} \quad y^{\star} = \begin{cases} x^2/4 & \text{if } x \geqslant 0, \\ -x^2/4 & \text{if } x < 0. \end{cases}$$

Example 9: Discuss this non-uniqueness

Existence and uniqueness

Example 7 (revisited): Reconsider the example from the start of the lecture

$$-xy' + y = 0, \quad \text{such that} \quad y(x_0) = y_0.$$

Here

$$f(x, y) = \frac{y}{x}.$$

f and f_y are unbounded when $x = 0$ and so our choice of rectangle R must avoid that line. So the theorem does not apply to cases such as $y(0) = 2$ and $y(0) = 0$.

We can employ the theorem for $y(1) = 2$ – where we do find a unique solution. Provided we choose a rectangle that avoids $x = 0$.

Existence and uniqueness in practice

To take advantage of this theorem we don't really need to think too much about rectangles and so on.

In practice, for an IVP $y' = f(x, y)$ with $y(x_0) = y_0$ just ask yourself two questions:

- for what region is $f(x, y)$ are continuous and bounded? (existence)
- within that region is f_y continuous and bounded? (uniqueness)

Having established that region, a unique this is the case solutions will exist provided the initial point (x_0, y_0) is within that region.

Further MATLAB - plotting functions

Example 10

Recall the hormone level example from earlier. We found the below solution by hand and by computer algebra.

$$y(t) = \frac{A}{K}\left(1 - \mathrm{e}^{-\kappa t}\right) + \frac{B}{\kappa^2 + \omega^2}\left[\kappa(\cos\omega t - \mathrm{e}^{-\kappa t}) + \omega\sin\omega t\right] + y_0\mathrm{e}^{-\kappa t}$$

Making a plot was slightly difficult because we had to input this expression by hand.

MATLAB can also directly plot symbolic functions, which is often more convenient.

Example 5 (revisited): use `fplot` to plot symbolic functions

Further MATLAB - numerical solution of algebraic equations

Not all equations (algebraic, DEs) have an analytic solution (ie. an answer you an write down).

Example 11

Consider the algebraic equation:

$$x = \cos x$$

(Ie. where do the functions x and $\cos x$ intersect?)

This has no analytic solution, but we can **approximate** the root numerically.

Example 10: Solve this with `fzero`

Further MATLAB - numerical solution of ODEs

Similarly we can use MATLAB to numerically solve differential equations.

Example 12

Consider the non-linear IVP

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y\cos(ty) + \cos t, \quad y(0) = 0.$$

This has no analytic solution, but we can find an **approximate** numerical solution for $y(t)$.

Example 11: Solve this IVP with `ode45`

Topic 1 summary

In the first topic we:

1. revised some techniques/ideas from Maths IB:
   - Modelling physical systems with differential equations
   - Solving separable ODEs
   - Solving linear ODEs (integrating factor)
2. introduced the use of MATLAB for:
   - Plotting functions (with `plot` and `fplot`)
   - Computer algebra for solving ODEs
   - Numerical method for solving algebraic equation (`fzero`)
   - Numerical solution of ODEs (`ode45`)
3. discussed **existence and uniqueness** of the solutions to first-order IVPs